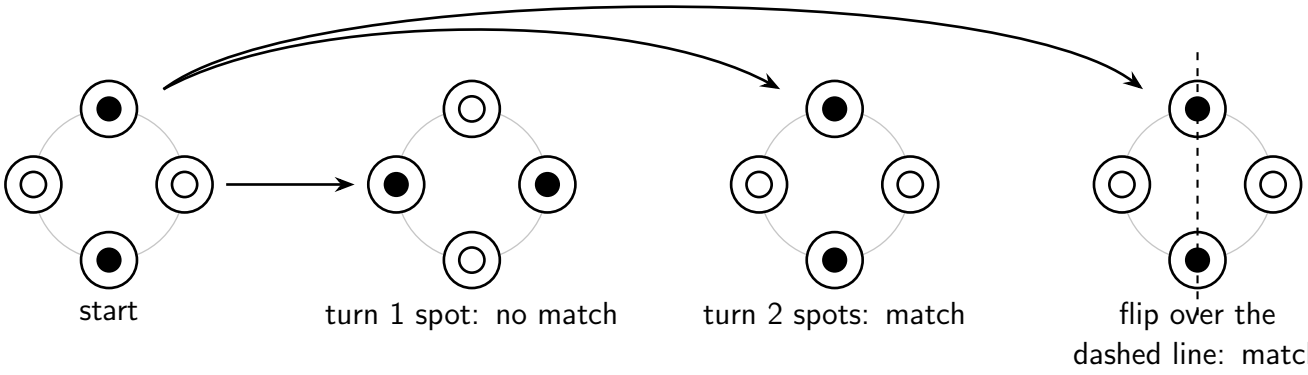
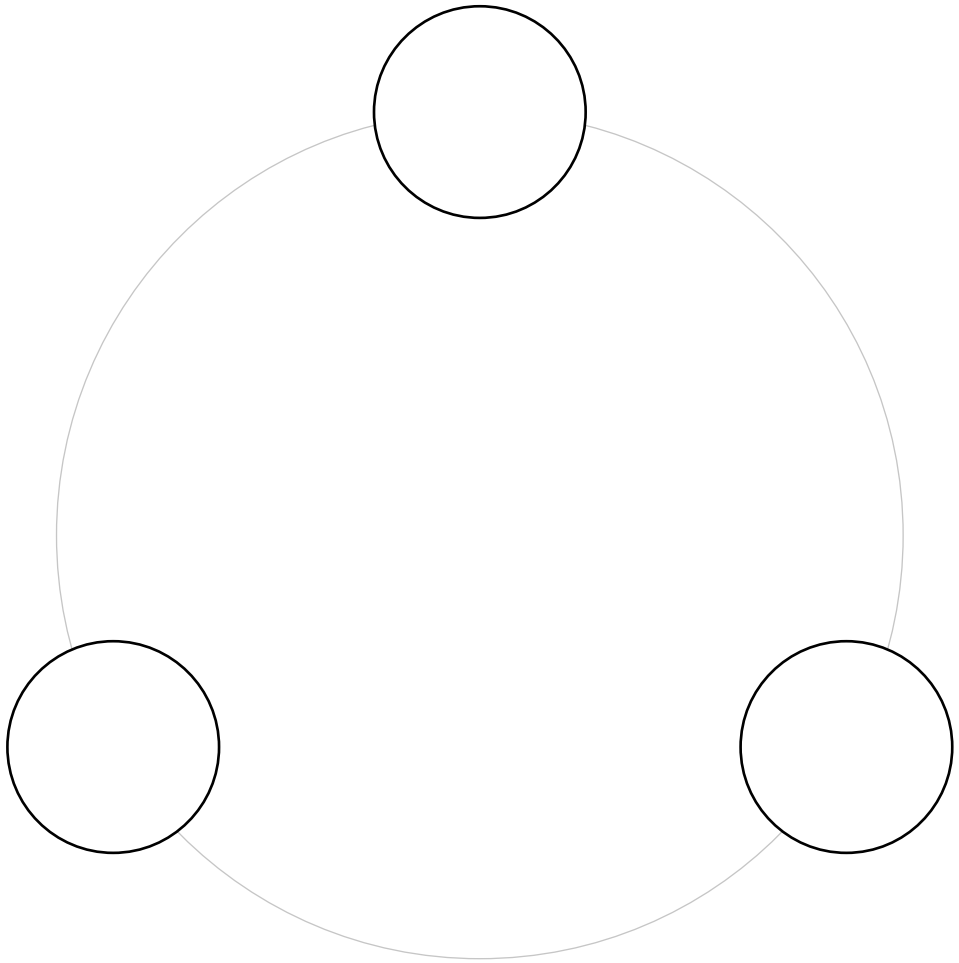
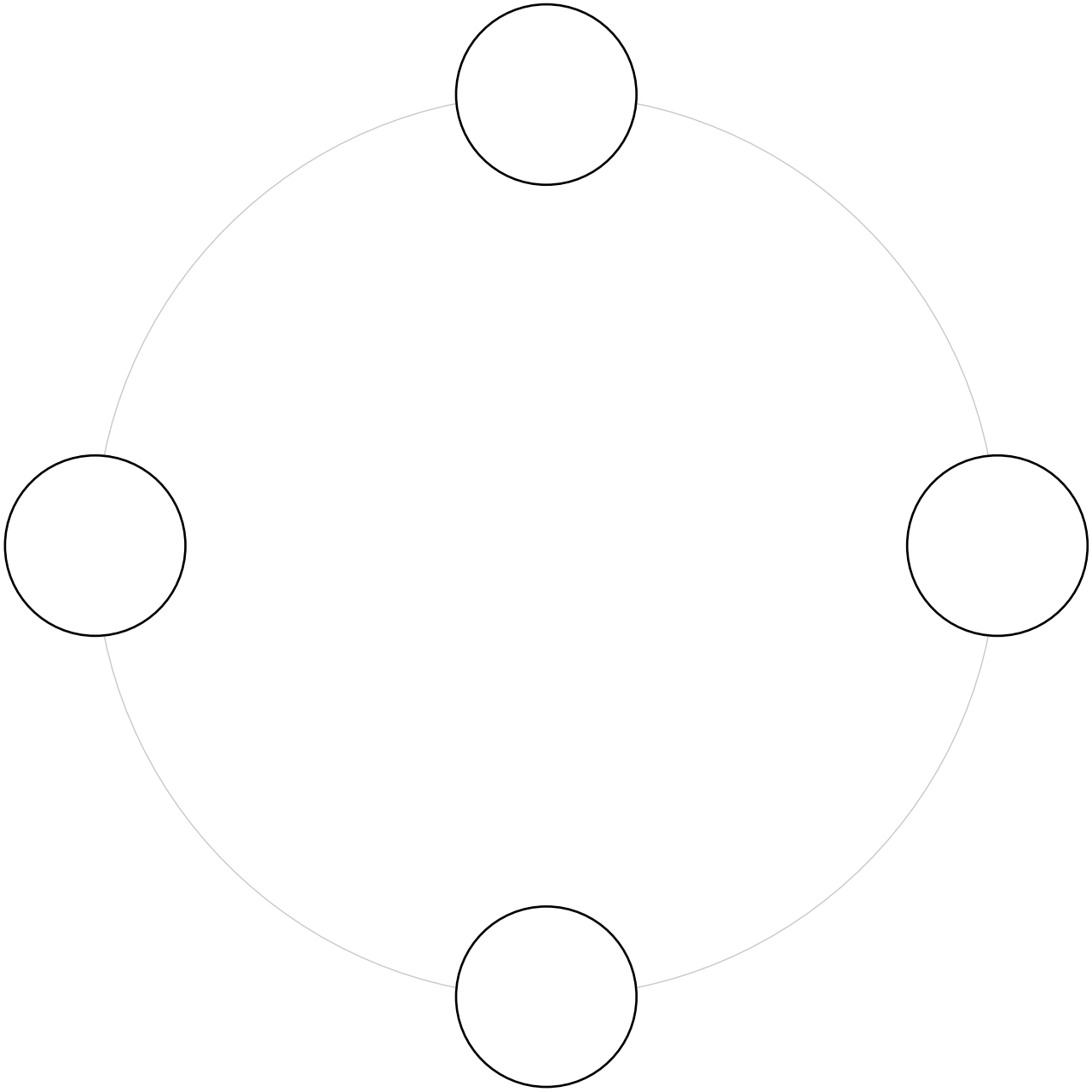
Fill every spot; neighbors may be the same kind. Only counter kinds and spots count. Copy a pattern on tracing paper with a dot of each counter's color. One partner builds. The other keeps the original still and turns or flips (turns over) the copy to make every dot sit on its own color. A whole turn does not count.



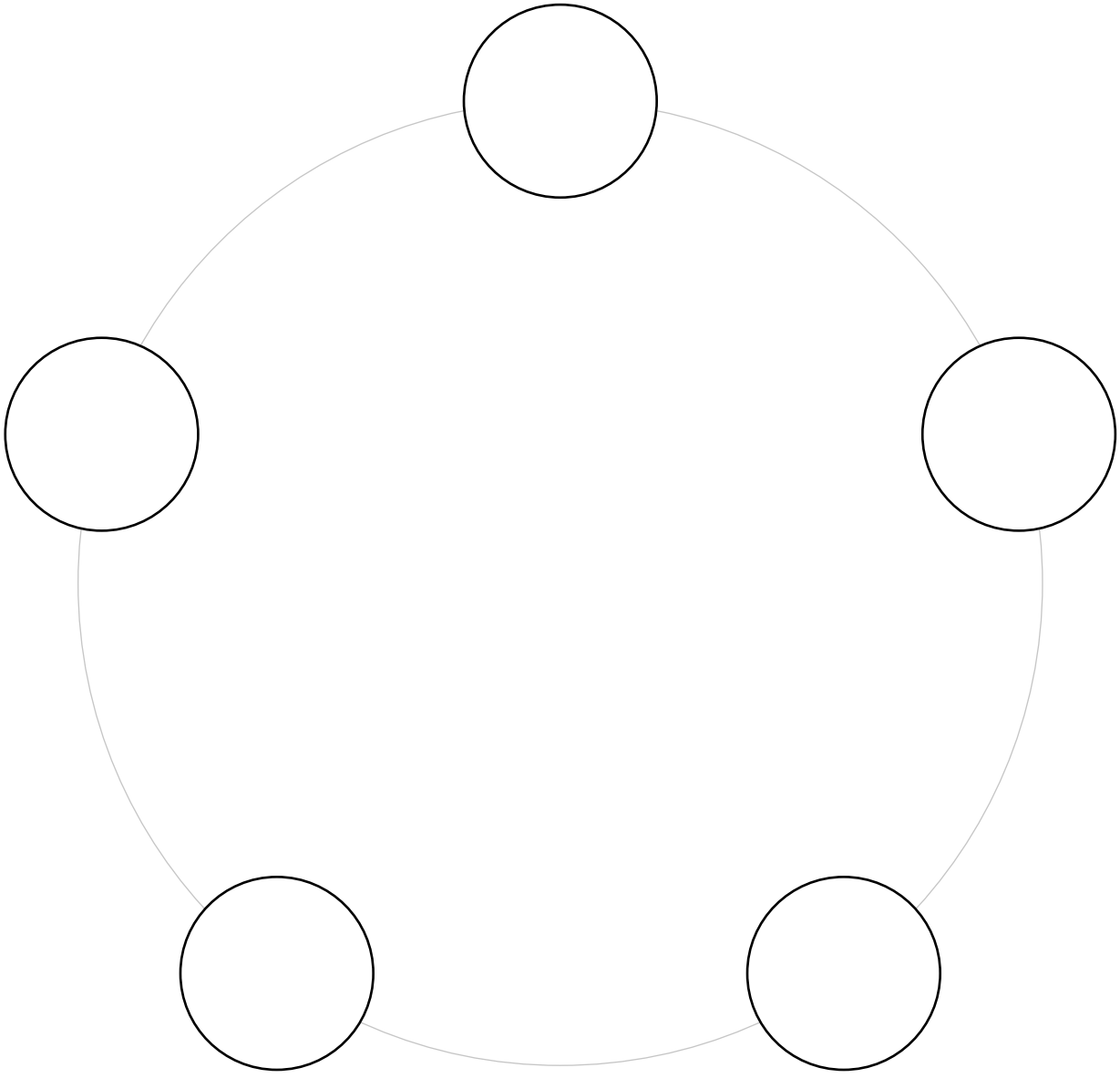
Problem 1: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match.



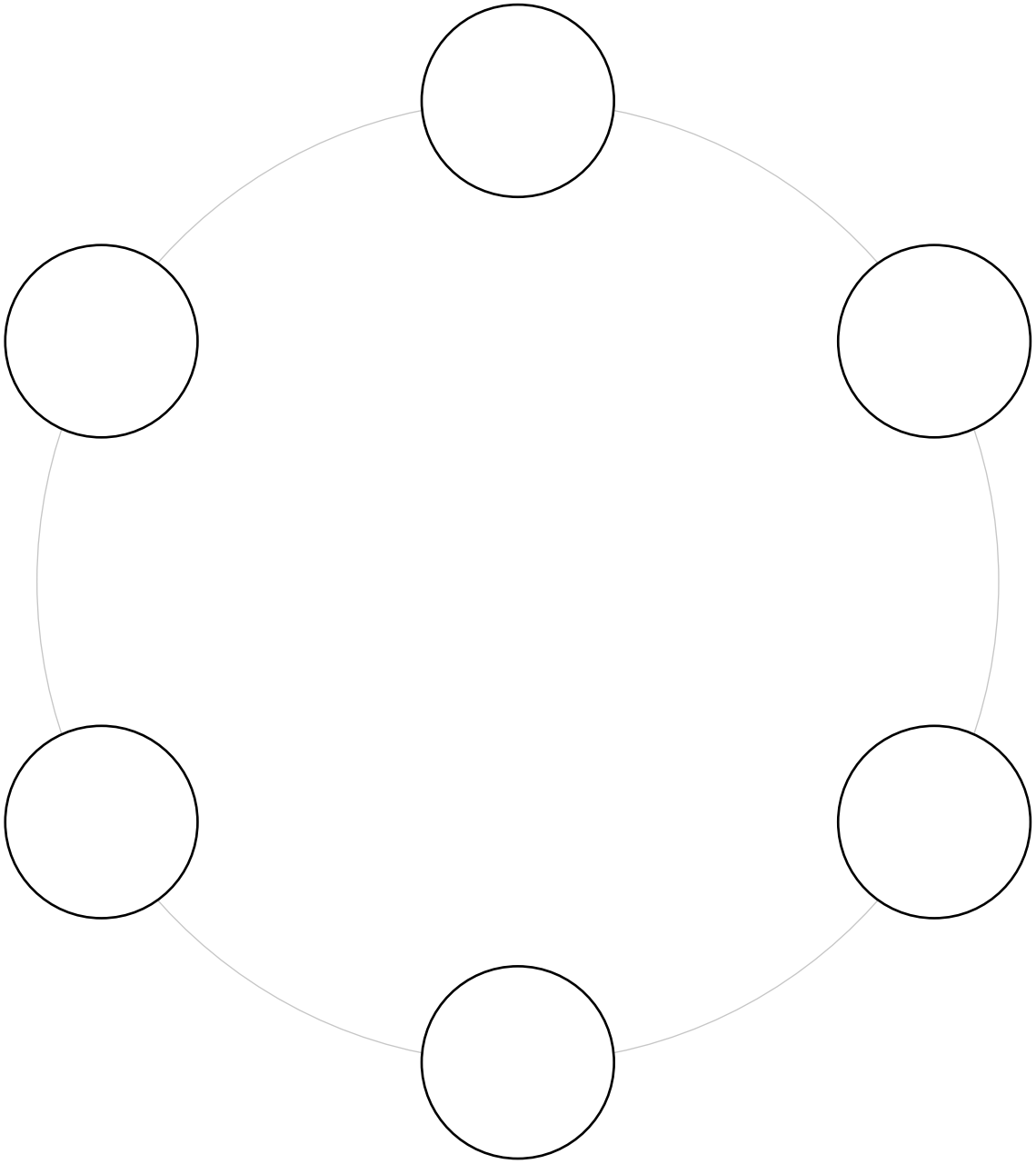
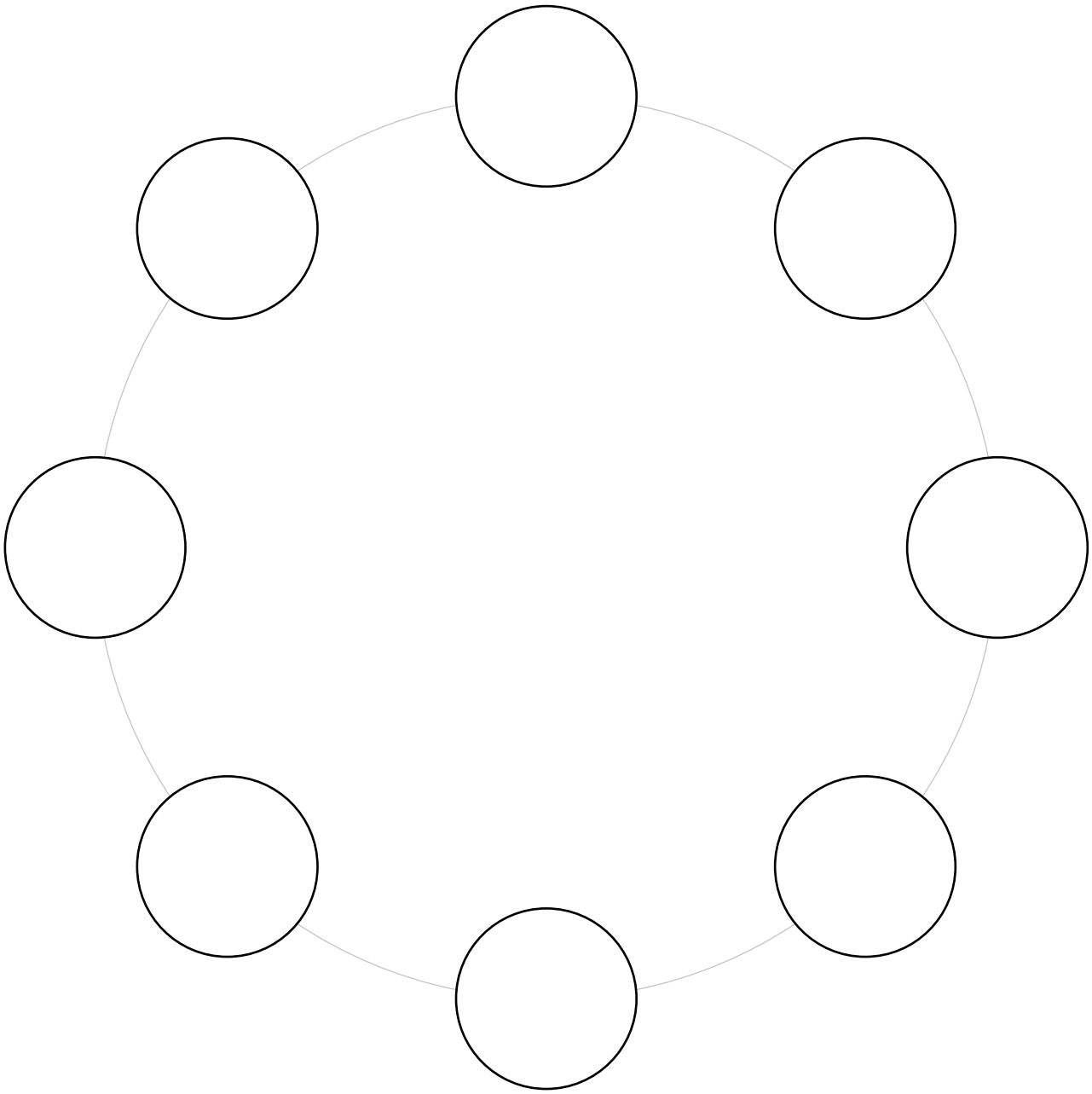
Problem 2: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match.



Problem 3: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match. Can you explain why fewer kinds would fail?



Problem 4: Find a six-ring pattern using only two kinds of counter that no turn or flip can match. Explain how you know no turn or flip can match it.



Problem 5: Find an eight-ring pattern with two kinds of counter that no turn or flip can match. How few counters of one kind can it have?